

Critical Design Review: Vehicle Control Unit Architecture for 1F2R Independent Hub Motor Auto-Rickshaw

Introduction and System Context

The transition from a conventional rigid rear axle to an independent dual rear hub motor architecture in a delta footprint (1F2R) commercial three-wheeler represents a profound shift in vehicle dynamics, control theory, and functional safety requirements. The proposed Vehicle Control Unit (VCU) logic relies heavily on advanced mathematical estimation techniques, including Unscented Kalman Filters (UKF), Recursive Least Squares (RLS) algorithms, and Second-Order Sliding Mode (SOSM) control. The primary objective of this architecture is to replicate the natural torque equalization of a mechanical differential while actively generating stabilizing yaw moments through differential motor torque. This must be accomplished under the stringent constraints of commercial operations, which include massive payload variations, uneven road gradients, and highly variable tire-road friction coefficients.

While the theoretical architecture is ambitious and attempts to address the core challenges of 1F2R torque vectoring, a rigorous mathematical, physical, and hardware-level analysis reveals critical vulnerabilities in the proposed solution. The evaluation indicates that the system architecture relies on idealized kinematic assumptions that do not hold under the nonlinear, transient dynamic conditions typical of a commercial auto-rickshaw operating near its physical limits. Furthermore, the discrete-time execution of the proposed control laws on automotive-grade microcontrollers introduces severe stability risks, and the degraded-mode fallback strategies violate both the thermodynamic limits of the electrical hardware and the fundamental principles of tire friction saturation.

This document serves as a comprehensive Critical Design Review (CDR), executed from the perspective of advanced control theory and ISO 26262 functional safety compliance. The review systematically deconstructs the mathematical, kinematic, and hardware frameworks of the proposed VCU, isolating the theoretical flaws and physical contradictions. For each identified vulnerability, a highly technical, mathematically rigorous correction is proposed to ensure the architecture satisfies the fundamental laws of vehicle dynamics and the strict safety mandates required for commercial deployment.

Mathematical and Kinematic Validations

The foundational physics engine of the VCU must accurately represent the highly asymmetrical and coupled dynamics of a delta three-wheeler. Unlike four-wheeled vehicles, the 1F2R topology presents unique challenges regarding load transfer, roll centers, and yaw coupling. The proposed mathematical models exhibit several critical oversimplifications that compromise the integrity of the predictive safety layers and the reference tracking models.

Rollover Logic and Delta Tipping Geometry

The predictive rollover algorithm relies on the Vectorial Rollover Equation ($PLTR_{delta}$) to anticipate un-tripped wheel lift before the chassis physically rolls. The VCU calculates a tipping angle defined as $\alpha = \arctan(L/(B/2))$, where L represents the vehicle wheelbase and B represents the rear track width. It then utilizes the projection equation $PLTR_{delta} = \frac{gd_{tip}}{h} (a_y \cos \alpha + a_x \sin \alpha) + C_p(\dot{\delta}_f v_x)$ to quantify the rollover threat.¹

While the geometric formulation of the angle α accurately describes the normal vector to the diagonal tipping axis relative to the longitudinal plane, the physical application of this projection equation is critically flawed under combined trail-braking and cornering scenarios. The fundamental error lies in the assumption that the vehicle's center of gravity (CG) remains statically positioned on the geometric centerline and at a constant vertical height h , and that the orthogonal lever arm d_{tip} is a fixed geometric constant. In reality, commercial auto-rickshaws possess a high center of gravity and relatively soft suspension setups designed for passenger ride comfort. During a high-G cornering maneuver, the chassis undergoes significant roll (ϕ), which displaces the CG laterally by a distance of $\Delta y = h \sin(\phi)$.² This lateral migration shifts the mass closer to the outer tipping axis, drastically reducing the required overturning moment. Furthermore, under trail-braking, longitudinal deceleration (a_x) induces suspension dive, which alters the longitudinal position of the CG (l_f), shifting the mass forward toward the single front wheel and further narrowing the stability triangle.²

The static orthogonal lever arm d_{tip} utilized in the proposed equation does not account for this dynamic spatial migration of the mass. Consequently, the equation will severely underestimate the true rollover propensity when the vehicle is simultaneously braking and turning. The physical restoring moment provided by gravity is diminished as the mass migrates outward and forward, meaning a lower combined acceleration threshold will trigger a rollover than the static equation predicts. The predictive derivative term $C_p(\dot{\delta}_f v_x)$ acts as a rudimentary phase-lead compensator but cannot mathematically replace the missing nonlinear kinematic terms governing the migrating center of mass. Relying on this formulation will result in the Danger Mode intervention triggering after the point of no return, failing to prevent un-tripped wheel lift.

Identification	Failure Analysis	Proposed Technical Correction
Slide 5: Vectorial Rollover	Fails to account for the dynamic spatial migration of	Implement a dynamic spatial

Equation ($PLTR_{delta}$)	the center of gravity ($\Delta y =$ and Δx) during chassis roll and suspension dive, leading to severe underestimation of rollover risk under combined acceleration.	$PLTF$ that incorporates real-time roll angle (ϕ) and pitch angle (θ) feedback. The lever arm must be formulated as a dynamic variable $d_{tip}(\phi, \theta)$ to account for the shrinking stability margin as the chassis deflects toward the tipping axis.
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Reference Model Fidelity for Single-Front-Wheel Kinematics

To define the driver's rotational intent, the architecture utilizes a dynamic reference model to

calculate the target yaw rate, expressed as $\omega_{des} = \frac{v_x \delta_f}{L + K_{us} v_x^2}$, where K_{us} is the understeer gradient.¹

This formulation is fundamentally derived from the standard four-wheeled, single-track (bicycle) model, which relies on the assumption of a lumped front axle and an open or mechanically linked rear differential.⁶ Applying this specific mathematical model to a 1F2R delta trike with independent rear hub motors introduces profound steady-state errors and unhandled slip angles.

The front axle of a delta trike consists of a single tire, which exhibits vastly different lateral force saturation characteristics compared to a lumped pair of tires on a four-wheeled vehicle.

Because a single front wheel cannot generate a roll restoring moment (there is no front anti-roll bar or track width to transfer load), the rear axle bears the entirety of the lateral load transfer

during cornering.⁶ This severely alters the physical understeer gradient K_{us} in a highly nonlinear manner as lateral acceleration increases.

More critically, the proposed vehicle utilizes torque vectoring via the independent rear hub

motors. The active generation of a differential yaw moment $M_z = \frac{(T_R - T_L)B}{2r_{tire}}$ creates a direct feedforward influence on the vehicle's yaw dynamics. The standard bicycle model assumes the longitudinal forces do not generate a primary yaw moment.⁷ By using an unmodified bicycle model to generate the reference yaw rate, the VCU creates a severe control paradox: the reference model dictates a trajectory based on a passive, un-driven rear axle, while the physical vehicle is actively driven by differential torque.

This mathematical mismatch guarantees that the target yaw rate will continuously diverge from the vehicle's physical capabilities. The Second-Order Sliding Mode (SOSM) controller will be forced to endlessly chase an mathematically impossible reference state. To minimize the

calculated error, the controller will demand excessive corrective torque (ΔT), forcing the vehicle into an artificially induced oversteer condition, thereby inducing permanent

steady-state slip angle errors at the front tire and degrading the natural driving feel demanded by the competition parameters.

Identification	Failure Analysis	Proposed Technical Correction
Slide 4: Target Yaw Rate Equation (ω_{des})	Utilizes a 4-wheel passive-axle bicycle model that completely ignores the direct yaw moment (M_z) generated by the independent rear hub motors, ensuring the reference model diverges from the physical vehicle's kinematic reality.	Formulate a dedicated 3-wheel (tricycle) reference model that incorporates the rear track width (B) and includes a steady-state feedforward term for the commanded differential longitudinal slip forces in the denominator of the understeer gradient.

Gradient Estimation and Suspension Pitch Decoupling

The architecture utilizes an Unscented Kalman Filter (UKF) to estimate the true road angle (θ_{road}) by decoupling it from the vehicle's suspension pitch. The proposed pitch-compensated gradient equation is $\theta_{road} = \arcsin\left(\frac{ga_{IMU} - \dot{v}_x}{g}\right) - K_{pitch} \cdot \hat{m} \cdot \dot{v}_x \cdot h_{LUT}$.¹

The critical flaw in this formulation lies in the treatment of the suspension pitch as an instantaneous algebraic gain (K_{pitch}). The physical suspension of an auto-rickshaw is a second-order mass-spring-damper system governed by the differential equation $I_y \ddot{\theta}_{pitch} + C_{pitch} \dot{\theta}_{pitch} + K_{\theta} \theta_{pitch} = M_{pitch}$.¹⁰ During the heavy stop-and-go traffic

conditions typical of commercial operations, the longitudinal acceleration ($\dot{v}_x$) is highly transient, characterized by frequent, high-jerk impulses as the driver applies and releases the throttle.

Under transient jerk, the vehicle chassis does not settle into a steady-state pitch angle instantaneously. Instead, it oscillates at its natural pitch frequency, damped by the shock absorbers. By applying a linear multiplier K_{pitch} to the instantaneous acceleration $\dot{v}_x$, the equation mathematically assumes that the suspension instantly reaches steady-state deflection the exact millisecond torque is applied. This creates a severe phase error between the estimated pitch and the true physical pitch of the chassis.¹¹

Consequently, the UKF will interpret the transient suspension oscillations as rapid, physical fluctuations in the road gradient. If the vehicle is attempting an uphill launch using the

proposed Anti-Rollback hold logic ($T_{hold} = \hat{m}g \sin(\theta_{road})r_{tire}$), these fictitious gradient fluctuations will destabilize the torque calculation. The hub motors will surge and stutter as they attempt to counteract a rapidly oscillating phantom gravity vector, violating the requirement for a natural and predictable driving feel.

Identification	Failure Analysis	Proposed Technical Correction
Slide 7: UKF Pitch-Compensated Gradient Equation	Models dynamic suspension pitch as an instantaneous, static linear gain (K_{pitch}), completely ignoring the second-order phase lag and transient oscillations of the physical mass-spring-damper system.	Integrate a dynamic pitch state (θ_{pitch}) directly into the UKF transition matrix, governed by the second-order differential equation of the suspension, to mathematically isolate transient chassis oscillations from the low-frequency road gradient.

Control Theory Practicality and Limit Handling

The translation of continuous-time, nonlinear control mathematics into discrete-time embedded C code on an automotive microcontroller exposes the system to a distinct class of vulnerabilities. The proposed limits handling and actuator allocation logic exhibit severe risks regarding mathematical continuity and non-minimum phase dynamics.

Discretization of the Super-Twisting Algorithm

To drive the kinematic yaw error (s) precisely to zero and eliminate the chattering typically associated with first-order sliding mode control, the VCU utilizes the Super-Twisting Algorithm (STA), a Second-Order Sliding Mode (SOSM) control technique. The corrective torque output is formulated as $\Delta T = -\lambda_1 |s|^{1/2} \text{sgn}(s) - \lambda_2 \int \text{sgn}(s) dt$.

While the STA theoretically guarantees a mathematically continuous control signal in the continuous-time domain, the reality of implementing this algorithm on an automotive-grade MCU operating at a 100-200Hz loop time introduces a catastrophic failure mode known as "discretization chattering".¹⁴ When the integral term $-\lambda_2 \int \text{sgn}(s) dt$ is digitized using a standard forward Euler or Runge-Kutta explicit numerical integration method, the state variable is updated discretely as $v_{k+1} = v_k - \lambda_2 T_s \text{sgn}(s_k)$.

Because the switching function $\text{sgn}(s_k)$ evaluates strictly to $+1$ or -1 , the controller outputs a discrete block of constant torque for the entire duration of the sampling period T_s (which ranges from 5 to 10 milliseconds). At 100-200Hz, this sampling period is massively too slow to contain the state trajectory within the ideal sliding manifold.¹⁶ The discrete control

effort will force the sliding variable s to continuously overshoot the origin, creating a

high-frequency, un-decaying limit cycle of order $\mathcal{O}(T_s)$.

This discretization chattering will manifest as violent 50Hz to 100Hz torque oscillations in the

independent hub motors. These oscillations will not only destroy the driving feel, inducing severe passenger discomfort, but will also induce severe mechanical fatigue, rapidly shearing the motor splines and degrading the stators through micro-vibrational thermal cycling. The claim that STA "completely eliminates rigid hub-motor chattering" is mathematically false when explicitly discretized at low frequencies.

Identification	Failure Analysis	Proposed Technical Correction
Slide 4: Corrective Torque Output (STA SOSM)	Explicit discrete-time integration of the discontinuous signum function at low loop frequencies (100-200Hz) mathematically guarantees high-frequency limit cycles (discretization chattering) that will physically destroy the actuators.	Replace the explicit Euler integration with an Implicit Discretization (Backward Euler) scheme for the STA, or replace the strict signum function with a boundary-layer saturation function tailored to the exact T_s of the MCU.

Friction Saturation Interlock and Topological Continuity

To manage actuator limits during extreme cornering and μ -split handling, the system relies on a Friction Saturation Interlock. The logic dictates that if the front tire slip (λ_f) approaches 1.0 (indicating a total loss of lateral grip and impending understeer plow), the target yaw moment (M_z) is instantly attenuated to zero to prevent a catastrophic spin-out.¹

This mechanism violates the foundational control theory requirement of topological continuity in torque allocation. Instantly zeroing the target yaw moment M_z equates to applying a Heaviside step function to the torque allocator. Consider a scenario where the vehicle is undergoing a high-G cornering maneuver and the outer rear motor is pushing heavily (positive torque) while the inner motor is dragging (negative torque) to vector the vehicle into the turn. If the front tire touches a patch of oil and λ_f spikes, an instantaneous drop of M_z to zero will cause the outer motor to instantly dump its positive torque, while the inner motor instantly dumps its braking torque.

This instantaneous release of differential torque will cause a massive and violent longitudinal load transfer shockwave across the rear axle.¹⁸ The sudden removal of the asymmetrical driving force will abruptly snap the vehicle's weight, generating an extreme jerk derivative. This shockwave will catastrophically overload the already saturated single front tire. Instead of mitigating the understeer, the instantaneous C^0 discontinuity in the actuator output will violently unsettle the chassis, resulting in snap oversteer or a highly aggressive un-tripped rollover. Control outputs must remain mathematically continuous to maintain chassis composure near the limits of adhesion.

Identification	Failure Analysis	Proposed Technical Correction
Slide 8: Friction Saturation Interlock	Instantly zeroing the target yaw moment creates a C^0 discontinuity (a mathematical step function) in the actuator commands, inducing a violent load transfer shock that will physically destabilize the vehicle.	Implement a C^1 continuous slip-based attenuation function (e.g., a sigmoid or exponential decay envelope) that smoothly blends the yaw moment to zero as $\lambda_f \rightarrow$, ensuring continuous derivatives in the torque commands.

Non-Minimum Phase Dynamics in Reverse Operations

To handle reverse driving and prevent jackknifing, the proposed architecture enforces a speed envelope that dynamically lowers the maximum reverse speed as the steering angle increases.¹ This solution demonstrates a fundamental misunderstanding of the kinematic reality of reversing a 1F2R vehicle. When driven in reverse, the delta trike topologically transforms into a rear-wheel-steered vehicle, operating under the exact same kinematic constraints as a forklift.²¹ Rear-steered vehicles are inherently unstable at speed because their lateral dynamics exhibit non-minimum phase behavior. Mathematically, the transfer function from steering input to lateral velocity contains a right-half-plane (RHP) zero.²²

Physically, a right-half-plane zero dictates that when the driver inputs a steering command, the rear of the vehicle initially swings violently in the *opposite* direction of the intended turn before the chassis eventually settles into the steady-state curve. A simple static envelope that caps speed based on steering angle completely fails to address this transient tail-swing. Even at extremely low speeds, the non-minimum phase dynamics dictate that any abrupt steering correction will cause the center of gravity to swing outward, initiating an uncontrollable pendulum effect that the driver cannot intuitively catch.

Because the standard forward-driving yaw reference model assumes front-wheel steering, feeding a reverse maneuver through the standard SOSM controller will result in positive feedback. The VCU will interpret the initial tail-swing as an error and apply differential torque that actively worsens the instability.

Identification	Failure Analysis	Proposed Technical Correction
Slide 8: Reverse State Logic Envelope	Fails to address the right-half-plane zero (non-minimum phase instability) inherent to rear-wheel-steered kinematics, where simple speed capping cannot prevent the violent	Design a dedicated inverted-kinematic control law specifically for reverse operation that actively utilizes the differential hub motors to provide aggressive, asymmetric yaw damping to

	transient tail-swing.	digitally suppress the non-minimum phase tail-swing.
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Operational Case Handling and Condition Coverage

Commercial three-wheelers operate in highly chaotic environments characterized by massive payload variations, heavy stop-and-go traffic, and degraded infrastructure. The mathematical estimators meant to handle these variations contain critical convergence and operational flaws that prevent adequate condition coverage.

Recursive Least Squares Convergence and Persistent Excitation

To dynamically map the true vertical CG height (h) for the rollover safety layer, the architecture relies on a Recursive Least Squares (RLS) algorithm to estimate total mass strictly from frequent, longitudinal stop-and-go accelerations.¹ In system identification theory, the RLS algorithm requires a strict condition known as "Persistent Excitation" (PE) to guarantee that the covariance matrix converges to the true parameter values.²⁵ Mathematically, the input regressor vector must possess sufficient spectral richness—a high degree of variance in acceleration and torque—to decouple vehicle mass from other confounding resistance forces, such as aerodynamic drag and minor road gradients. In heavy, creeping stop-and-go traffic, an auto-rickshaw rarely exceeds 10-15 km/h, and the longitudinal acceleration profiles are low-amplitude, low-frequency, and heavily polluted by sensor noise and road irregularities. These conditions entirely lack the necessary Persistent Excitation.²⁶ If three heavy passengers board the vehicle, shifting the total mass from 400 kg to 900 kg, the RLS algorithm will remain frozen at the previous 400 kg estimate until a sufficiently aggressive and sustained acceleration maneuver occurs to satisfy the PE condition. If the driver immediately pulls out of a stop and executes a sharp, high-speed evasive maneuver to merge into traffic before the RLS has converged, the VCU will calculate the $PLTR_{delta}$ rollover index using the obsolete 400 kg mass and its correspondingly low CG height. The Danger Mode intervention will trigger far too late, allowing the 900 kg vehicle to physically roll over before the software recognizes the elevated threat level.

Identification	Failure Analysis	Proposed Technical Correction
Slide 5: Payload Variance & Dynamic CG Tracking	RLS mass estimation cannot converge rapidly in low-excitation stop-and-go traffic, leaving the rollover safety layer operating on obsolete, dangerously low mass estimates immediately following passenger loading.	Integrate a static mass estimation routine at zero-velocity utilizing initial motor startup torque versus micro-rotation, or suspension deflection mapping, to instantaneously re-initialize the RLS covariance matrix upon passenger loading.

Adhesion Limit Handling on μ -Split Surfaces

To handle μ -split surfaces, the architecture claims to calculate the front tire's maximum restoring capacity by pulling steering angle, estimated velocity, and payload data. However, the calculation of the pseudo-velocity reference $\hat{v}_x$ explicitly blends in the rear-wheel speeds (v_{RL} and v_{RR}) during front lockup, defined as $\hat{v}_x = \dots + W_r \cdot$

If the vehicle encounters a strict μ -split surface (e.g., the left wheel is on wet mud or ice, and the right wheel is on dry tarmac), the slip calculation $\lambda = \frac{\max(\omega r, \hat{v}_x) - \min(\omega r, \hat{v}_x)}{\max(\dots)}$ will be fundamentally corrupted. Because the rear axle lacks a mechanical differential, the application of longitudinal torque will cause the wheel on the low-friction surface to instantly over-speed, while the wheel on the high-friction surface maintains grip. If the UKF blindly utilizes the over-speeding wheel to calculate the baseline chassis velocity, the state estimator will artificially inflate the predicted vehicle speed $\hat{v}_x$. Consequently, the VCU will interpret the rotational speed of the gripping wheel as being vastly slower than the chassis speed, leading to a false calculation of massive negative slip (braking) on the high-friction side. The torque allocator will subsequently attempt to correct this phantom slip by dumping torque to the gripping wheel and sending more torque to the slipping wheel, inducing severe torque steer directly into the low-friction side and pulling the vehicle off the road.²⁹

Identification	Failure Analysis	Proposed Technical Correction
Slide 8: μ -Split Surfaces & TCS Slip Calc	Blending unfiltered independent rear wheel speeds into the UKF velocity reference on μ -split surfaces will corrupt the chassis speed estimate, inducing positive feedback torque steer.	Implement a variance-weighted kinematic observer in the UKF that dynamically isolates and rejects the rotational speed of any wheel exhibiting a sudden derivative spike ($\dot{\omega} \gg$ threshold) indicative of μ -split saturation.

Functional Safety (ISO 26262) and Hardware Degradation

The transition into fallback states during catastrophic sensor or hardware failure must guarantee the physical safety of the occupants and surrounding traffic. The proposed ASIL degradation states violate the physical limits of both the vehicle kinematics and the

electromagnetic capabilities of the permanent magnet synchronous motors (PMSM).

States 1-3: Kinematic Redundancy and Actuator Authority

In the event of IMU loss (State 2), the degradation matrix dictates that the VCU abandons active yaw feedback, enforces severe torque limits (under 20%), and strictly caps speed to initiate a 15 km/h limp-home mode.¹

Capping speed and torque does not neutralize the inherent danger of a dual independent hub motor architecture operating blindly. The fundamental physical danger of the 1F2R architecture without a mechanical axle is asymmetrical thrust. If the IMU is lost, the VCU has no closed-loop feedback to verify that the vehicle is traveling in a straight line. If the left inverter experiences a minor hardware calibration fault, or if the left tire is slightly under-inflated (reducing its rolling radius), a command of 20% maximum torque to both motors will result in unequal longitudinal forces at the contact patches.²⁹

Even at a capped speed of 15 km/h, a constant asymmetrical application of torque on a short-wheelbase, 900 kg vehicle will generate a slow but relentless yaw moment. Without the IMU to detect this uncommanded rotation, the vehicle will steadily plow out of its lane and into oncoming traffic. Relying merely on low speed does not replace the requirement for kinematic synchronization between the two physically disconnected actuators.

Identification	Failure Analysis	Proposed Technical Correction
Slide 9: State 2 (IMU Loss) Fallback	Speed and torque caps do not prevent uncommanded yaw moments caused by asymmetrical motor performance, tire radius variance, or independent inverter faults when operating blindly without IMU feedback.	The Torque Allocator must default to a strict "Virtual Locked Axle" mode, forcing the left and right wheel speed encoders to track identically ($\omega_L =$) via localized PI control, overriding all torque requests to guarantee straight-line travel.

State 6: The Electro-Hydraulic Brake Blending Paradox

If the brake pedal sensor fails but the hydraulic lines remain intact (State 6), the VCU enforces "One-Pedal Drive" (heavy lift-off regenerative braking) for electronic deceleration while retaining full mechanical braking capability.¹

This strategy introduces a fatal human-machine interaction conflict. Under normal operation, drivers rely on the brake pedal for primary deceleration. If a hazard appears, a driver operating in the failed sensor state will instinctively lift completely off the throttle and immediately slam their foot onto the mechanical brake pedal. Because the VCU's brake sensor is dead, it cannot detect that the driver is physically applying the hydraulic brakes.

Consequently, the VCU will simultaneously apply maximum lift-off regenerative braking torque to the rear hub motors at the exact millisecond the driver applies maximum hydraulic friction

braking to the exact same wheels. The superposition of maximum electronic braking torque and maximum mechanical braking torque will instantly exceed the longitudinal friction limits of the rear tires.³⁰ The rear axle will lock up entirely, reducing the lateral cornering stiffness of the rear tires to zero. At high speeds, rear-wheel lockup guarantees an immediate and violent spin-out, transforming a simple sensor fault into a fatal loss of vehicle control.

Identification	Failure Analysis	Proposed Technical Correction
Slide 10: State 6 (Brake Sensor Death)	Enforcing heavy lift-off regen while mechanical hydraulics are intact guarantees instantaneous rear-wheel lockup if the driver panics and applies both braking forces simultaneously.	Completely disable heavy regenerative braking if the brake pedal sensor fails. Rely exclusively on a gentle coast-down map, forcing the driver to rely purely on the intact mechanical hydraulics for all deceleration to prevent overlapping torque saturation.

State 7: Thermodynamics of Pulsed Active Short Circuits

In the event of a total primary VCU collapse (State 7), the architecture relies on an independent System Basis Chip (SBC) and a hardware astable multivibrator to pulse the Active Short Circuit (ASC) pins at a fixed PWM duty cycle of 15% ON / 85% OFF. The stated intent is to shed kinetic energy without exceeding thermal limits.¹

This hardware failsafe is physically disastrous from both an electromagnetic and thermodynamic perspective. An Active Short Circuit relies on turning on all three high-side or low-side IGBTs/MOSFETs to short the motor's back-EMF, creating a braking torque bounded by the stator resistance and inductance. If the ASC is pulsed via a 15% PWM duty cycle, the inverter must continuously interrupt massive fault currents. Because the primary VCU is dead, there is no active Space Vector modulation, no current control loop, and no dead-time management.³¹ Interrupting the highly inductive current of the PMSM stator 10,000 times a second using a dumb hardware multivibrator will generate catastrophic voltage spikes (

$V = L \frac{di}{dt}$) that will instantly obliterate the power electronics through avalanche breakdown.

Furthermore, the physics of a 900 kg vehicle on a 15% downgrade dictate a continuous

gravitational force of $F = mg \sin(\theta) \approx 900 \cdot 9.81 \cdot 0.15 = 1324 \text{ N}$. A 15% duty cycle implies that the vehicle is freely accelerating for 85% of every switching period. The RMS braking force generated by a 15% pulsed ASC will be vastly insufficient to counteract the 1324 N continuous gravitational pull.³⁵ The vehicle will exponentially accelerate down the grade while simultaneously destroying its inverter hardware.

Identification	Failure Analysis	Proposed Technical Correction
Slide 10: Thermal-Safe	Pulsing the ASC via dumb	Abandon PWM modulation.

Deceleration (Duty-Cycled ASC)	hardware will trigger catastrophic inductive voltage spikes, destroying the inverter, while the 15% duty cycle provides insufficient RMS torque to stop a 900kg vehicle on a 15% downgrade.	Implement a hard-latched, continuous (100% duty cycle) three-phase symmetric ASC. A continuous ASC inherently limits terminal velocity by balancing back-EMF with stator impedance and generates zero switching losses, preventing hardware destruction.
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Conclusion

The proposed Vehicle Control Unit architecture for the 1F2R auto-rickshaw incorporates highly sophisticated control methodologies, but relies on theoretical approximations that collapse under real-world vehicle dynamics and embedded hardware constraints. The failure to account for dynamic mass migration in the rollover index, the incompatibility of the reference model with active torque vectoring, and the discretization chattering of the Super-Twisting Algorithm present severe risks to vehicle stability and hardware integrity.

Furthermore, the ISO 26262 degradation strategies exhibit fatal flaws regarding kinematic redundancy, human-machine interaction, and electromagnetic thermodynamics. To achieve a stable, predictable, and safe vehicle that meets the demanding operational parameters of the Bajaj Campus Challenge, the VCU physics models must be fundamentally revised to embrace the nonlinear kinematics of the 1F2R topology, and the control algorithms must be heavily constrained by the physical realities of discrete-time microcontrollers and thermodynamic boundaries. Implementations of the technical corrections detailed in this review are mandatory to prevent catastrophic loss of vehicle control during extreme limit handling and hardware fault conditions.

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